Solutions

1. See separate sheet.

2. See separate sheet.

3. a) i) The auxiliary polynomial is $a(\lambda) = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1)$ and so the complementary function is $\boxed{y_c(x) = c_1 e^x + c_2 e^{-x}, \quad c_1, c_2 \text{ constants.}}$

   ii) Use a trial solution $y_p(x) = b_1 + b_2 x e^x$ in the differential equation and equate coefficients to get $b_1 = b_2 = 1$ and so $\boxed{y_p(x) = xe^x + 1.}$

   iii) The general solution is then $y(x) = c_1 e^x + c_2 e^{-x} + xe^x + 1$. Enforcing the initial conditions gives $c_1 = -1$ and $c_2 = 0$ and so $\boxed{y(x) = xe^x - e^x + 1.}$

   b) i) Write the differential equation as $P(x,y)dx + Q(x,y)dy = 0$. The equation is exact if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \Leftrightarrow \quad \lambda_1 x - \cos x \sin y = 2x - \frac{1}{2}\lambda_2 \cos x \sin y \Leftrightarrow \boxed{\lambda_1 = \lambda_2 = 2.}$$
$$\Rightarrow \quad (2xy + \cos x \cos y)\,dx + \left(x^2 - \sin x \sin y\right) dy = 0.$$

   ii) Since $df = f_x dx + f_y dy$ we need $f_y = Q$ and so

$$\frac{\partial f}{\partial y} = x^2 - \sin x \sin y \Rightarrow \boxed{f(x,y) = x^2 y + \sin x \cos y}$$

   iii) Since $df = 0$ the solution is given implicitly by $f(x,y) = C$ or

$$\boxed{x^2 y + \sin x \cos y = C, \quad C \text{ constant.}}$$

   c) i) The integrating factor $\mu(x)$ must be chosen to satisfy

$$\frac{d\mu(x)}{dx} = \mu(x)\,\frac{3}{x} \Rightarrow \boxed{\mu(x) = \exp\left(\int \tfrac{3}{x}\,dx\right) = e^{3\ln x} = x^3.}$$

   ii) Multiplying the equation by $\mu(x)$:

$$x^3 \frac{dy}{dx} + 3x^2 y = 2x \quad \Rightarrow \quad \frac{d}{dx}\left(x^3 y\right) = 2x$$
$$\Rightarrow \quad y(x) = x^{-3} \int 2x\,dx + Cx^{-3}$$
$$\Rightarrow \quad \boxed{y(x) = x^{-1} + Cx^{-3}, \quad C \text{ constant}}$$

4. a) i) Evaluating the partial derivative for the chain rule

$$\begin{bmatrix} \frac{\partial f}{\partial x} \\ \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{x}{\sqrt{x^2+y^2}} & \frac{-y}{x^2+y^2} \\ \\ \frac{y}{\sqrt{x^2+y^2}} & \frac{x}{x^2+y^2} \end{bmatrix} \begin{bmatrix} \frac{\partial f}{\partial \rho} \\ \\ \frac{\partial f}{\partial \phi} \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi/\rho \\ \sin\phi & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \frac{\partial f}{\partial \rho} \\ \\ \frac{\partial f}{\partial \phi} \end{bmatrix}$$

and so

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = \begin{bmatrix} \frac{\partial f}{\partial \rho} & \frac{\partial f}{\partial \phi} \end{bmatrix} \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi/\rho & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi/\rho \\ \sin\phi & \cos\phi/\rho \end{bmatrix} \begin{bmatrix} \frac{\partial f}{\partial \rho} \\ \frac{\partial f}{\partial \phi} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial f}{\partial \rho} & \frac{\partial f}{\partial \phi} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/\rho \end{bmatrix} \begin{bmatrix} \frac{\partial f}{\partial \rho} \\ \frac{\partial f}{\partial \phi} \end{bmatrix} = \left(\frac{\partial f}{\partial \rho}\right)^2 + \frac{1}{\rho^2}\left(\frac{\partial f}{\partial \phi}\right)^2$$

and so $\boxed{a = 1.}$

ii) Since f is radially symmetric, $f_\phi = 0$ and the equation transforms to

$$\left(\frac{\partial f}{\partial \rho}\right)^2 = \rho^2 \Rightarrow \frac{\partial f}{\partial \rho} = \pm\rho \Rightarrow f(\rho) = \pm\frac{1}{2}\rho^2 + C \Rightarrow \boxed{f(x,y) = \pm\tfrac{1}{2}(x^2+y^2) + C.}$$

b) i) Since $dF = 0$ we have that $\frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy + \frac{\partial F}{\partial z}dz = 0$. It follows that

$$\frac{\partial z}{\partial x} := \frac{dz}{dx}|_{y=\text{constant}} = \frac{dz}{dx}|_{dy=0} = -\frac{\partial F}{\partial x}\Big/\frac{\partial F}{\partial z} \Rightarrow \boxed{\frac{\partial z}{\partial x} = \frac{2x}{z}.}$$

and since F is symmetric in x and y, $\boxed{\frac{\partial z}{\partial y} = \frac{2y}{z}.}$

ii) The stationary points are obtained by setting $z_x = z_y = 0$ and so $\boxed{x_0 = y_0 = 0}$ from Part (i). The same answer could be obtained by explicitly expressing z explicitly as a function of x and y. Note that $z(x_0, y_0) = 2$.

iii) Taking the second derivatives and evaluating at x_0, y_0,

$$\frac{\partial^2 z}{\partial x^2}|_{x_0,y_0} = \frac{\partial}{\partial x}\left(\frac{2x}{z}\right)|_{x_0,y_0} = \frac{2z - 2xz_x}{z^2}|_{x_0,y_0} = 1 = \frac{\partial^2 z}{\partial y^2}|_{x_0,y_0}$$

For the mixed derivatives,

$$\frac{\partial^2 z}{\partial x \partial y}|_{x_0,y_0} = \frac{\partial}{\partial x}\left(\frac{2y}{z}\right)|_{x_0,y_0} = \frac{-2yz_x}{z^2}|_{x_0,y_0} = 0$$

The Hessian is then

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $m_{11} > 0$ and $\det M > 0$, the stationary point corresponds to a $\boxed{\text{local minimiser.}}$